Intrinsic Motivation for Noise-Robust Exploration

From MiniWorld coverage to sparse-reward MiniGrid — a full response to `docs/SPEC.md`

Challenging Problems in Reinforcement Learning (SS 2026)

Topic: Exploration vs. Exploitation — reproducing and extending Hou, An & Du (2026), LPM

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Abstract

These notes cover the project end to end — both halves of `docs/SPEC.md`. **Part I (SPEC §2, slide deck §2) reproduces the LPM paper** on its own 3D **MiniWorld** maze: 6 methods $\times$ 3 noise variants $\times$ 64 seeds, A2C, the paper's 5×10^4-step budget, with a paper-faithful log-space LPM. The headline noisy-TV robustness claim *reproduces* (under `action_noise` the raw-pixel predictor MSE fixates on the noise wall 83% of the time and its coverage collapses, while LPM almost never looks at it, 3%); but the paper's "LPM achieves the highest coverage" ranking does *not* — a *uniform-random* control covers the most in all three variants. The lesson: **coverage is the wrong yardstick in an extrinsic-free maze**, which motivates Part II. **Part II (SPEC §3, slide deck §3) is our extension to sparse-reward MiniGrid**, where a real task reward lets intrinsic motivation pay off and *task success* discriminates exploration quality. Under a common PPO agent (MLP on the 7×7 symbolic view) we compare a plain baseline (NONE), entropy-bonus exploration (ENTROPY), Random Network Distillation (RND) and the paper's Learning Progress Monitoring (LPM) across a three-rung difficulty ladder (easy DoorKey-5x5, medium FourRooms, hard MultiRoom-N6), clean and under observation noise. Three results. **(1)** The intrinsic coefficient β matters enormously and is environment-dependent: β=0.05 *drowns* the sparse reward; the usable range is ∼0.001–0.005. **(2)** Intrinsic motivation is *difficulty-gated*: unnecessary on easy/medium, but **decisive on hard MultiRoom-N6, where RND solves the task (eval 0.32) while baseline/entropy never reach the goal**. **(3)** That trade-off reverses under noise: on noisy DoorKey-5x5 **RND becomes unstable while LPM and the baseline stay robust**, and a fine FourRooms noise sweep (0–0.1) confirms the separation is **RND-vs-the-rest**: only RND degrades steeply, while NONE/ENTROPY/LPM stay tied. So LPM reproduces the paper's *noise-robustness* (it avoids RND's noise-chasing collapse) but, in this extrinsic-reward domain, is noise-*neutral* — it does not beat a plain baseline that is already robust. Figures, tables and agent traces below are mapped to the planned slide deck.

Part I — Reproducing the LPM paper on MiniWorld (slide deck §2)

1 MiniWorld environment and setup (slides 2.1–2.2)

Environment. The paper's exploration testbed is a hand-designed 3D **MiniWorld** maze (four rooms, 160×120 RGB first-person observations) with *no extrinsic reward* — the agent is scored purely by how much of the maze it visits. Two "noisy-TV" variants stress-test robustness: `noisy_tv`, where one wall shows white noise (a green-pixel → random-RGB transform), and `action_noise`, where a dedicated action (#4) replaces the whole observation with a random CIFAR-10 image — an unlearnable, perpetually "novel" distractor that a prediction-error bonus will chase forever. `nonoise` is the clean control.

Agent, methods, metrics. A2C (the upstream pipeline), shared CNN encoder, 5×10^4 exploration steps per run — the paper's maze budget (Fig. 3 of Hou et al.). We compare **six** policies: LPM (paper-faithful log-space learning-progress reward $r^i = g_\phi - \log \mathrm{MSE}$), RND, ICM, MSE (raw next-frame pixel error, the pure noisy-TV victim), NONE (A2C, no intrinsic reward), and a *uniform-random* control ($a \sim \mathrm{Uniform}\{0 \ldots 4\}$, no learning — the "does any of this beat chance?" baseline). Metrics: **coverage** (fraction of reachable cells visited), **TV-action share** (fraction of steps spent on the noise action under `action_noise` — the direct fixation measure), and **coverage-heatmap evolution** (per-cell occupancy over training). *We made LPM paper-faithful: log-space Eq. 3 instead of the released notebooks' raw-error clipped form, which silently masked a stability issue. Full design + numbers:* `latex_notes/2026-05-31-maze-exploration-design.pdf`*; slides:* `reports/report2/`.

2 Result 1 — noisy-TV robustness reproduces (slide 2.3)

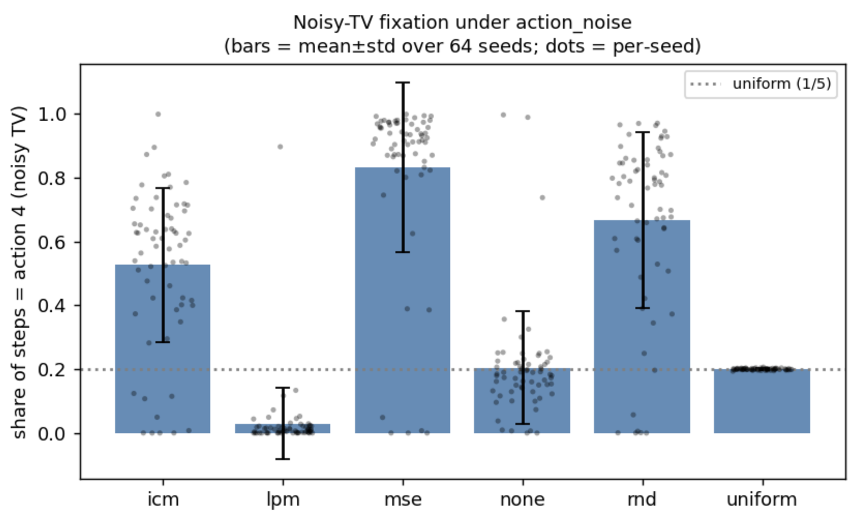


Figure 1: TV-action share under `action_noise` (bars = mean±std; black dots = the 64 per-seed values): fraction of steps spent staring at the noise wall. MSE is trapped on it (0.83); LPM essentially never is (0.03, below even the no-learning baseline); the curiosity baselines RND/ICM are partially fooled; *uniform* sits at exactly 1/5 (chance). The per-seed dots show the full distribution behind each bar — LPM is a tight mass at ≈ 0, while MSE/RND/ICM spread high (each with a low-tail of lucky seeds), confirming the effect is not a mean-only artifact.

The mechanism is exactly the paper's: a prediction-*error* bonus (MSE) rewards the perpetually surprising noise wall and the agent fixates; LPM rewards model *improvement*, which is 0 on unlearnable noise, so LPM ignores the wall. RND and ICM, which also reduce to prediction-error signals, are partially fooled (67%, 53%). *Slide 2.3.*

Policy	TV-action share (`action_noise`)
LPM	$\mathbf{0.030 \pm 0.113}$
uniform	0.200 ± 0.003
NONE	0.205 ± 0.175
ICM	0.526 ± 0.242
RND	0.666 ± 0.276
MSE	$\mathbf{0.830 \pm 0.266}$

Table 1: Noisy-TV fixation, 64 seeds. LPM almost never stares at the noise wall (median 0.9%; only 1/64 seeds ever fixates) because its dynamics model cannot *improve* on the unlearnable image; MSE is trapped (83%). **This is the paper's central claim, and it reproduces.**

3 Result 2 — but coverage does not separate methods (slides 2.4–2.5)

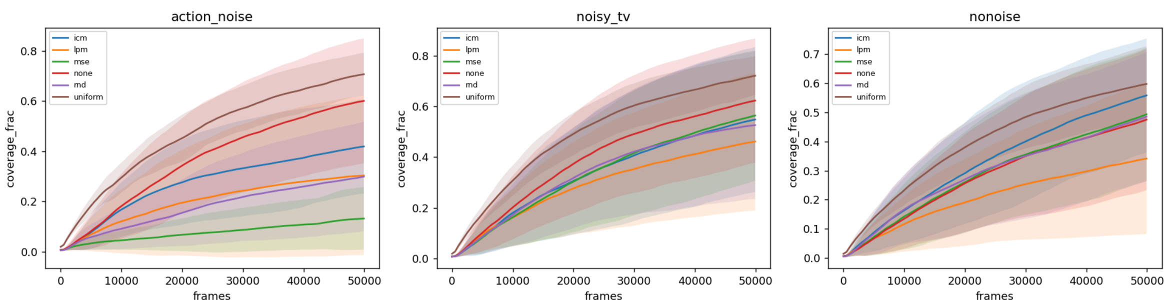


Figure 2: Coverage vs. frames, per variant (mean±std, 64 seeds). The *uniform-random* control (brown) covers the most in all three variants and with the lowest variance; every *learned* policy — LPM included — falls below it. Bands are wide (between-seed std 0.08–0.32): the single-env A2C is high-variance.

Conclusion: coverage is not a good exploration metric (slide 2.5). In this small, extrinsic-free maze, coverage rewards undirected *randomness* — uniform thrashing spreads through the rooms while any *learned* policy commits to a narrower, repetitive walk, so training *reduces* coverage. Noise-avoidance and coverage are decoupled (under `action_noise` uniform wastes 20% of steps frozen on the wall yet still covers most; LPM never touches the wall yet covers less on average). **So what *is* a good metric for exploration?** Our answer, and the pivot of the project: move to a setting with a real, sparse *task* reward, and measure *success at the task*. That is Part II. *Slide 2.5 — the bridge to the MiniGrid work.*

Policy	`nonoise`	`noisy_tv`	`action_noise`
uniform	**0.587 (0.13)**	**0.710 (0.08)**	**0.696 (0.09)**
NONE	0.459 (0.24)	0.611 (0.25)	0.589 (0.25)
LPM	0.332 (0.25)	0.452 (0.27)	0.299 (0.32)
MSE	0.476 (0.23)	0.551 (0.26)	0.129 (0.12)
ICM	0.542 (0.20)	0.536 (0.29)	0.410 (0.19)
RND	0.468 (0.22)	0.519 (0.29)	0.290 (0.22)

Table 2: Coverage fraction, mean (std), 64 seeds. Best per column in **bold**: the uniform-random control wins *all three* variants. **The paper's "LPM covers the most" ranking does not reproduce here.** LPM's coverage is further depressed by a bimodal seed collapse (policy instability of the unclipped λ_i=1 reward, *not* TV-fixation — the collapsed seeds freeze somewhere in the maze, not on the wall).

Part II — Intrinsic motivation in sparse-reward MiniGrid (slide deck §3)

4 Setup

Observation space. A MiniGrid step returns a dictionary {`image`, `direction`, `mission`}. The `image` is the agent's *egocentric, partially-observable* 7×7 view of the cells *in front of it* (not the full grid), with **3 channels** per cell encoding (object id, colour id, state) in MiniGrid's compact *symbolic* scheme — e.g. object$\in$ {wall, door, key, goal, ...}, state$\in$ {open, closed, locked} for doors. These are *not* RGB pixels. We wrap with `ImgObs` (keep only the image) and flatten to a **147-dim** (= 7×7×3) vector. *This answers "why were the observations 2835-dim?": the default* `FlatObs` *appends a one-hot encoding of the mission string to the image, and that mission is fixed per task — so ∼95% of* `FlatObs` *is a constant, uninformative padding.* `ImgObs` *carries the identical spatial information ∼19× lighter; we use it throughout.*

Action space. MiniGrid's native action space is Discrete(7): {0 : turn left, 1 : turn right, 2 : move forward, 3 : pickup, 4 : drop, 5 : toggle/open, 6 : done}. We restrict it to the minimal useful subset *per environment* (a `MiniGridActionSubsetWrapper`), which removes dead actions and shrinks the exploration branching factor: **FourRooms** (pure navigation) uses {left, right, forward} (**3** actions); **DoorKey** and **MultiRoom-N6** (must grab a key / open doors) use {left, right, forward, pickup, toggle} (**5** actions); the never-needed `drop` and `done` are dropped.

Model and PPO hyperparameters. PPO (Stable-Baselines3), `MlpPolicy` (the SB3 default: separate actor and critic MLPs, two hidden layers of 64 with tanh), **8** parallel envs. Rollout `n_steps`=512 per env (⇒ a 8×512=4096-transition buffer per update), `batch_size`=64, `n_epochs`=10, `learning_rate`=2.5e−4, γ=0.99, GAE λ=0.95, `clip_range`=0.2, `ent_coef`=0 (the ENTROPY arm overrides this to 0.01). Deterministic evaluation every 5k steps over 10 episodes (each a freshly randomized layout; see *Procedural layout randomization* below) on a clean-of-intrinsic copy of the env. The intrinsic reward enters the *training* return as $r = r^{\text{ext}} + \beta\, r^{i}_{\text{norm}}$; eval return is the pure sparse task reward.

Exploration methods. (i) NONE: plain PPO, ent_coef=0. (ii) ENTROPY: PPO with ent_coef=0.01 — the *non-intrinsic* (policy-stochasticity) comparator. (iii) RND: bonus = $\|f_\theta(s) - \hat{f}(s)\|^2$ (predictor vs. fixed random target). (iv) LPM: paper-faithful log-space reward $r^i = g_\phi - \log \mathrm{MSE}$, gated until the error buffer fills.

Environments (difficulty ladder, one per tier). easy `DoorKey-5x5` (subgoal shaping disabled → purely sparse), medium `FourRooms`, hard `MultiRoom-N6`. *KeyCorridor-S3R3 was tried as the hard rung and abandoned: it requires* memory *(carry a key, remember it, open a door), which a feed-forward MLP on the image-only view cannot represent — all methods scored* 0 *even at 3M steps. MultiRoom-N6 is memory-free (doors just toggle) so it is MLP-solvable, but exploration-hard.*

Procedural layout randomization (every episode). All three environments *regenerate their layout on every reset* — during training *and* during evaluation; the random seed only fixes the RNG stream, not a single fixed maze. The tiers differ in *how much* they vary. `DoorKey-5x5` is only mildly random: the structure is fixed (agent in the left room, one door in the dividing wall, key in the left room, and the **goal pinned in the far corner**), with just the key/door/spawn cells jittering by a step or two. `FourRooms` places **both the spawn and the goal uniformly at random** across the four rooms, so a single episode can be trivial (same room, a few cells apart) or demand crossing all four. `MultiRoom-N6` draws a fresh 6-room chain each episode (random room sizes, placements and door positions) with the goal in the final room — always far from the spawn. Consequently **every reported eval return is an average over this layout distribution**, not a single-maze score: this is the representative way to measure generalisation, and it is why FourRooms' mean sits low (it averages a wide spread of easy and hard goal placements). *A fixed-seed render therefore shows only* one *draw from this distribution: the agent-trace film-strips (§8) are single representative episodes, chosen for clarity rather than as summaries of the layout distribution. The caveat bites hardest for FourRooms, where a fixed seed can land a trivial same-room layout; DoorKey/MultiRoom are safer because their goal is* structurally *far from the spawn (fixed far corner / end of the room chain).*

Observation noise. The noisy variants corrupt the symbolic image with a wrapper whose `noise_prob` is the per-*cell* corruption probability: each of the 7×7=49 cells of the egocentric view is independently corrupted with probability `noise_prob` (so "10% noise" means ≈ 5 of the 49 cells), and a corrupted cell is re-drawn as a unit with each of its three channels resampled *within its valid range* (object $\in [0, 10]$, colour $\in [0, 5]$, state $\in [0, 2]$, read from MiniGrid's constants). This is a *global* distractor (any cell can be hit), not a localised noisy-TV wall, and it never injects encodings the agent could not otherwise see. *Corrected from an earlier per-*element *model that flipped each of the* 147 *channel-values independently and replaced them with out-of-range values — that made "10%" touch* ≈ 27% *of cells, so the noisy numbers below supersede earlier drafts.*

Metrics, seeds, budget. Final deterministic eval return (the sparse task reward; the eval env carries no intrinsic bonus), the per-episode extrinsic/intrinsic split, and sample-efficiency curves. Mean±std over seeds: **8 seeds on DoorKey-5x5** (bumped up to resolve LPM's high variance there; see §7), **3 seeds** elsewhere. Budgets: easy/medium 1M, hard 2M PPO steps.

5 β: how to mix extrinsic and intrinsic reward (RQ2)

At β=0.05 the per-episode *intrinsic* reward was 100–700× the sparse extrinsic reward, so the policy optimises novelty and never learns the goal (eval → 0). The usable band is ∼0.001–0.005. On the *hard* env a slightly larger β is needed to drive exploration, but too large still prevents convergence to exploitation: RND on MultiRoom-N6 only converges at β=0.005 (eval 0.28); at β=0.01/0.05 it reaches the goal during training but never settles into a reliable deterministic policy (eval ≈ 0). **Takeaway:** β is environment-dependent — small when intrinsic reward is a nuisance, larger when it is essential, with a sweet spot in between. *Slide 3.3.*

6 Does intrinsic reward beat the baseline? (RQ3)

env (clean)	NONE	ENTROPY	RND	LPM
DoorKey-5x5 (easy)	0.94	0.91	0.87	$0.67^{\dagger}$
FourRooms (medium)	0.34	0.30	0.30	0.25
MultiRoom-N6 (hard)	0.00	0.00	**0.32**	0.00

Table 3: Final eval return (mean). **Intrinsic motivation is difficulty-gated.** DoorKey-5x5 over **8 seeds**, the rest over 3. †LPM-clean DoorKey is *bimodal* (6/8 seeds solve ≈ 0.96, 2/8 collapse to 0; mean 0.67, std 0.41) — see §7.

Intrinsic motivation's benefit appears *only* when exploration is genuinely hard. On easy and medium tasks the PPO baseline explores adequately on its own and any bonus slightly degrades it; on hard MultiRoom-N6 the baseline and entropy never reach the goal, while **RND solves it reliably** (0.32, all seeds). *Slide 3.4.*

Why RND > LPM on clean hard exploration. Per-episode intrinsic magnitudes show the mechanism: RND's bonus is *always positive* (mean = abs), a consistent directional drive toward novel states; LPM's bonus is *signed* and nets ≈ 0 per episode (mean +0.08 vs. abs 0.33), giving no coherent exploration push — so the LPM agent behaves like the baseline and never explores out. Raising β scales LPM's oscillation amplitude, not its directionality, so no β rescues it. This is LPM's designed trade-off: its learning-progress signal goes to 0 both for unpredictable noise (its robustness virtue, next section) *and* for already-learned local dynamics (so it under-explores clean hard tasks).

7 Noise robustness: LPM vs. RND (RQ4)

DoorKey-5x5	clean	noisy (10%)
NONE	0.94	0.96
ENTROPY	0.91	0.96
RND	0.87	$\mathbf{0.52}^{\ddagger}$
LPM	$0.67^{\dagger}$	**0.96**

Table 4: The cleanest noise-robustness result (**8 seeds**, clean *and* noisy, under the corrected cell-level noise model). DoorKey-5x5 is small enough that 10% noise does not stop NONE/ENTROPY/LPM from solving — which *isolates* the intrinsic-reward noise-vulnerability: only RND is hurt. †LPM-clean is bimodal (6/8 solve $\approx$ 0.96, 2/8 collapse; mean 0.67, std 0.41); under noise LPM solves on all 8 seeds (std 0.001). ‡RND-noisy is highly variable (per-seed $\{0.03, 0.04, 0.20, 0.47, 0.50, 0.96, 0.96, 0.96\}$; mean 0.52, std 0.40) — noise destabilises it where LPM and the baseline are unaffected.

Observation noise makes every state look "novel," so RND's prediction-error bonus stays high and the agent chases noise instead of the goal; LPM's learning-progress signal stays near 0 on unlearnable noise, so LPM is not distracted and effectively reverts to the (working) base policy. DoorKey-5x5 (Table 4) isolates this cleanly because the task is easy enough that noisy perception alone does not break it — so only RND is hurt, and *unstably*: 3/8 seeds still solve (≈ 0.96) while 3/8 collapse (≤ 0.05), against LPM's 8/8. On FourRooms (next paragraph) the global noise degrades *every* method and RND separates as clearly the worst, while none/entropy/lpm stay tied (at 0.1: none 0.10, entropy/lpm 0.09, rnd 0.03). On hard MultiRoom-N6, noise breaks perception for everyone (RND's clean 0.32 $\rightarrow$ 0). *Slides 3.4/3.6.*

FourRooms: a fine noise sweep separates RND, but not LPM from the baseline. A natural worry about the single-point (0.1) comparison is that the noise is simply too strong, flattening any differences among none/entropy/lpm. To check, we swept `noise_prob` from 0 to 0.1 in steps of 0.01 (3 seeds; Fig. 6). The result is clarifying: **RND separates cleanly** — it degrades faster than every other method and pulls below the pack from ≈ 0.04 onward ($0.07 \rightarrow 0.03$) — but **none, entropy and LPM stay clustered and statistically tied** (pairwise gaps ≤ 0.02, within the 3-seed std) at *every* noise level. LPM never beats the plain baseline; at low noise it sits slightly *below* it (its clean under-exploration deficit) and only *catches up* as noise rises. *This corrects a tempting over-read: a per-point "% of clean return retained" makes LPM look most robust (38% vs.* entropy 31% *at* 0.1*), but that ratio is inflated by LPM's* lower clean baseline *— in absolute terms LPM* $\approx$ none $\approx$ entropy *throughout. The honest, consistent RQ4 conclusion across DoorKey and FourRooms:* only RND is noise-vulnerable; LPM is noise-neutral *— it avoids RND's noise-chasing failure but does not, on these extrinsic-reward MiniGrid tasks, confer an advantage over a plain PPO baseline (which, lacking an intrinsic bonus to be fooled, is itself robust).*

LPM-clean variance is bimodality, not sampling noise (8-seed check). The one place LPM's 3-seed band looked alarmingly wide was *clean* DoorKey-5x5. We bumped it to **8 seeds** to check whether more data would tighten it. It does not: the per-seed final returns are $\{0.97, 0.96, 0.97, 0.97, 0.97, 0.87\}$ for six seeds and $\{0.00, 0.00\}$ for two, i.e. LPM *either* solves the task (≈ 0.96) *or* under-explores and never reaches the goal (0), with no middle ground (mean 0.67, std 0.41). Adding seeds sharpened this picture rather than shrinking the std — the variance is *intrinsic bimodality*, not small-sample uncertainty. This is the same signature as the MiniWorld result (Part I): the faithful, unclipped λ_i=1 LPM reward is noise-robust but prone to a per-seed policy collapse. RND, by contrast, is *unimodal* on the same clean task (8/8 seeds solve, ≈ 0.87, std 0.08, *no* collapse) — its always-positive bonus gives a stable exploration drive, where LPM's net-zero signal can leave a seed stuck. (Under *noise* the roles partly invert: it is RND that turns unstable, §7.) *Reporting this honestly is the point; the wide band is a property of LPM, not a measurement defect.*

8 Agent traces (slide 3.5)

Animated GIFs live in `expr_data/minigrid/figures/traces/`; representative film-strips below.

9 Conclusions — answers to the research questions

RQ1 *What do the LPM-paper experiments say about exploration metrics?* Coverage (MiniWorld) is a poor metric in extrinsic-free settings (a uniform-random policy maximises it). In MiniGrid we use *task success / return*, which actually discriminates exploration quality.

RQ2 *Correct way to mix extrinsic/intrinsic; β too small/large?* β too large drowns the sparse reward ($\rightarrow$ 0); too small is ignored. Usable $\sim$0.001–0.005, environment-dependent; on hard tasks there is a sweet spot (≈ 0.005 for RND) between under-exploring and never-converging (Fig. 4).

RQ3 *Does intrinsic reward beat baselines on sparse MiniGrid?* Only when exploration is hard: **yes, decisively, on MultiRoom-N6 (RND** 0.32 **vs.** 0.0**)**; no on easy/medium where the baseline suffices (Fig. 5). RND > LPM on clean hard exploration.

RQ4 *Does LPM generalise to other noise (beyond the MiniWorld noisy-TV)?* Yes — **LPM is noise-robust and RND noise-vulnerable**, cleanest on DoorKey-5x5 (under noise LPM solves 8/8 at 0.96 while RND drops to 0.52 and turns highly unstable, per-seed 0.03–0.96; Table 4). A fine FourRooms sweep (0–0.1, Fig. 6) confirms the separation is *RND-vs-the-rest*: only RND degrades steeply, while NONE/ENTROPY/LPM stay statistically tied. So LPM reproduces the paper's robustness in the sense that it *avoids RND's noise-chasing failure*, but on these extrinsic-reward MiniGrid tasks it is noise-*neutral* — it does not beat a plain baseline (which, having no intrinsic bonus to be fooled, is itself robust).

One-line story for the deck: *intrinsic motivation buys you hard-exploration ability (RND) at the price of noise-fragility; LPM trades that exploration aggressiveness away to avoid the noise-fragility — so it never collapses like RND, but on these tasks it only matches the plain baseline rather than beating it* — and β must be tuned or it drowns the task.

Slide-deck mapping (full SPEC)

Deck §2 — Reproducing the LPM paper (Part I).

- 2.1 MiniWorld env / 2.2 model, setup & metrics → §1.
- 2.3 TV-action share → Fig. 1, Table 1.
- 2.4 coverage curve + 2.4 coverage heatmap (larger variance, LPM not best) → Figs. 2, 3, Table 2.
- 2.5 conclusion "coverage is not a good metric" → end of §3 (the bridge to Part II).

Deck §3 — Intrinsic reward and LPM (Part II).

- 3.1 environments / 3.2 model & setup → §4.
- 3.3 β sweep → §5, Fig. 4.
- 3.4 success rate across envs (clean and noisy) → §6–§7, Fig. 5, Table 3, Fig. 6.
- 3.5 trace demonstrations → §8, Fig. 7.
- 3.6 LPM vs. RND, clean and noisy gap → §6 (clean: RND>LPM) + §7 (noisy: LPM>RND) — the rank flip is the punchline.

Deck §4 — Conclusion → the four-RQ summary above and the one-line story.

A Reproducibility & data

Part I (MiniWorld). Code: `LPM_exploration/Miniworld/experiments/` (vendored from the authors' release, with the log-space LPM fix logged in `LPM_exploration/UPSTREAM.md`); raw position logs, figures and tables in `expr_data/miniworld/`; full design + per-cell numbers in `latex_notes/2026-05-31-`

and slides in `reports/report2/`. **Part II (MiniGrid).** Code: `minigrid_exp/` (vendored + extended from Youssef's `minigrid_intrinsic_reward`). Raw data, tables, figures and trace GIFs: `expr_data/minigrid/` with a running log in `FINDINGS.md`. **Compute note:** the 128-core box SIGKILLs any single process after ∼18 min, so long runs use chunked checkpoint–resume training (`train_one --chunk-steps`, resume via `PPO.load`) driven by a meta-loop re-invoking `run_grid.py` (`ThreadPoolExecutor`); each chunk finishes under the limit and the idle driver survives. Grids: `run_grid.py --envs ... --methods rnd lpm --betas ... --noise-probs ... --seeds 1 2 3`. **Caveats:** (a) the MiniGrid noise is *global cell-level* corruption (each of the 7×7 cells independently corrupted with prob. 0.1 and re-drawn within valid channel ranges), not a localised noisy-TV distractor; a localised variant would be a cleaner analogue. (b) LPM under-explores clean hard tasks here, so the clean LPM-vs-RND gap is RND-favourable; under noise LPM avoids RND's collapse but, on a fine FourRooms sweep, only *matches* the plain baseline rather than beating it (the robustness win is over RND, not over NONE). (c) Seeds: DoorKey-5x5 at 8 seeds *both clean and noisy* confirms LPM-clean's wide band is *genuine bimodality* (6/8 solve, 2/8 collapse), not small-sample noise; the noisy 8-seed cells re-ran cleanly under the corrected noise model. All other envs are at 3 seeds.

B Theoretical-max eval return

MiniGrid gives `reward` $= 1 - 0.9\,(\text{steps}/\text{max\_steps})$ on reaching the goal (0 on timeout), so the *theoretical-max* eval return (the eval averages 10 random layouts) is $1 - 0.9\,\mathrm{E}[d^*]/\text{max\_steps}$, where d^* is the optimal action count to the goal. We compute d^* exactly by breadth-first search over (position, direction, carrying, opened-doors), using the agent's own action repertoire (turn left/right, forward, pickup, toggle), and average the optimal reward over 300 random layouts per env (`minigrid_exp/optimal_reward.py`). Results:

env	`max_steps`	opt. steps	**theoretical-max return**
DoorKey-5x5	250	∼9.8	**0.965**
FourRooms	100	∼16.0	**0.856**
MultiRoom-N6	120	∼46.5	**0.652**

The ceiling is well below 1.0 (reaching the goal already costs a fraction of the step budget), and lower for the longer environments. This calibrates the result figures: on DoorKey the solvers run near-optimal (≈ 0.96 vs 0.965), whereas on FourRooms (∼0.3 vs 0.86) and MultiRoom-N6 (RND 0.32 vs 0.65) every method leaves large headroom — these tasks are far from solved, not near-saturated.

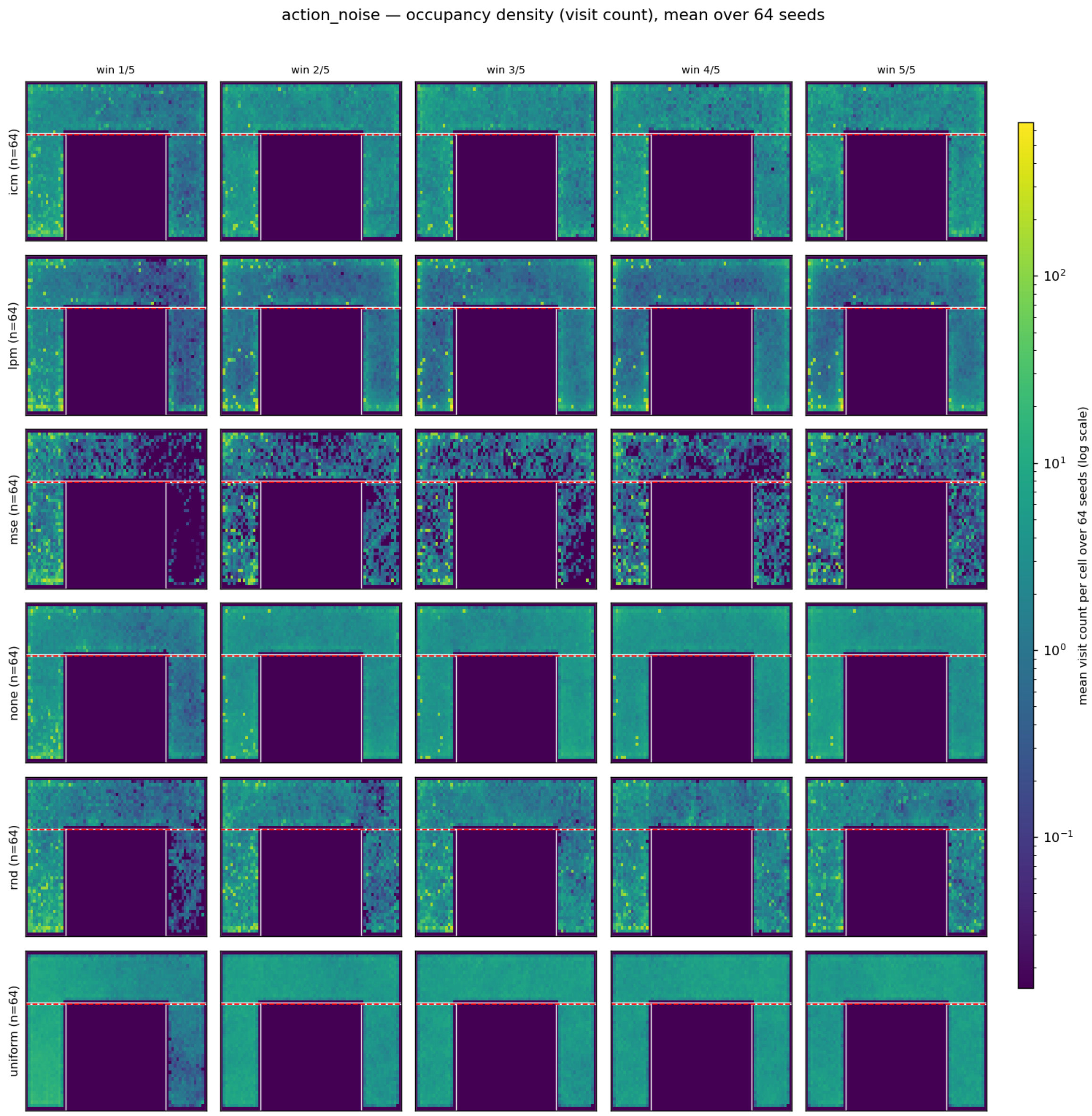


Figure 3: Coverage-heatmap evolution under `action_noise` (per-cell visit density, shared log scale, mean over 64 seeds; rows = method, columns = training windows). MSE's mass stays pinned at the noise wall (top strip) while the room empties; *uniform* fills the rooms most evenly. Visual confirmation that fixation (Table 1) and coverage (Table 2) are *decoupled*.

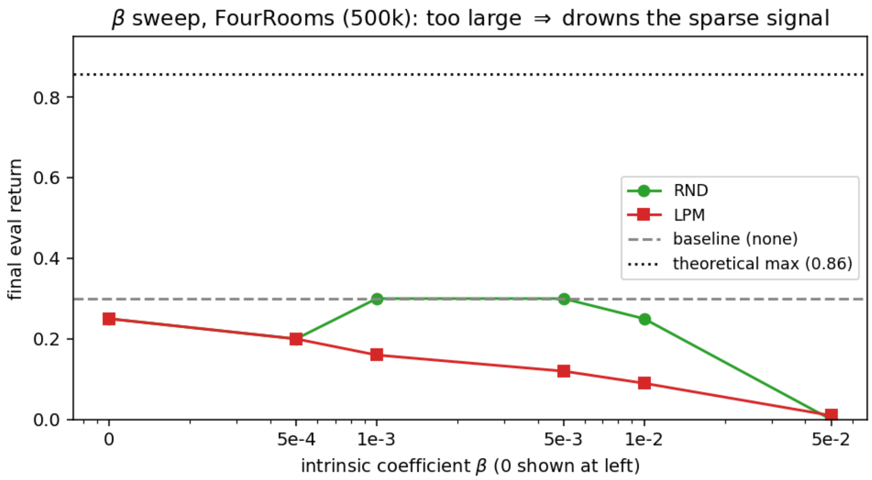


Figure 4: β sweep on FourRooms (500k). RND rises to a sweet spot ($\beta \approx 0.001$–0.005, matching the baseline) then collapses to 0 at β=0.05; LPM declines monotonically. The dotted line is the *theoretical-max* eval return for FourRooms (0.86, optimal play; see §B) — every method sits far below it, i.e. nobody solves FourRooms reliably.

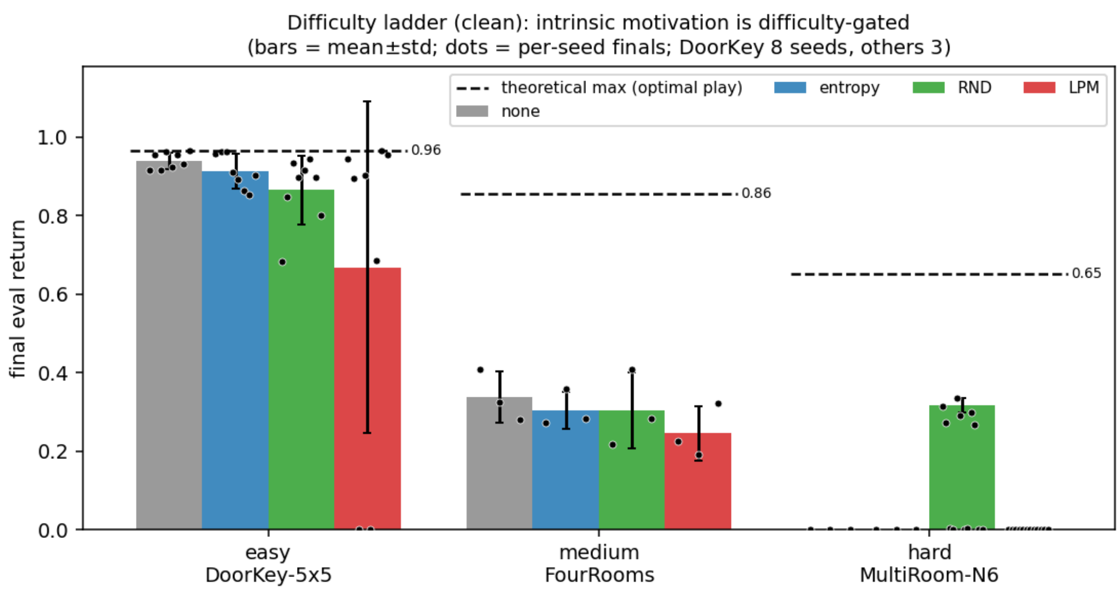


Figure 5: Difficulty ladder, clean, final eval return (bars = mean±std; black dots = the per-seed finals, 8 on DoorKey, 3 elsewhere — same windowed definition as the bars). Easy/medium: the baseline already solves/explores; intrinsic is unneeded (and slightly hurts). Hard MultiRoom-N6: only RND reaches the goal (its 3 dots sit at ≈ 0.32 while every other method's dots lie on the floor at 0). The dots also expose that LPM-clean DoorKey's wide bar is a 0/0.96 *bimodal split*, not a uniform spread. The dashed per-env lines are the *theoretical-max* eval returns (optimal play; §B): DoorKey 0.97, FourRooms 0.86, MultiRoom-N6 0.65 — so on DoorKey the solvers are near-optimal, whereas on FourRooms/MultiRoom everyone is far from the ceiling.

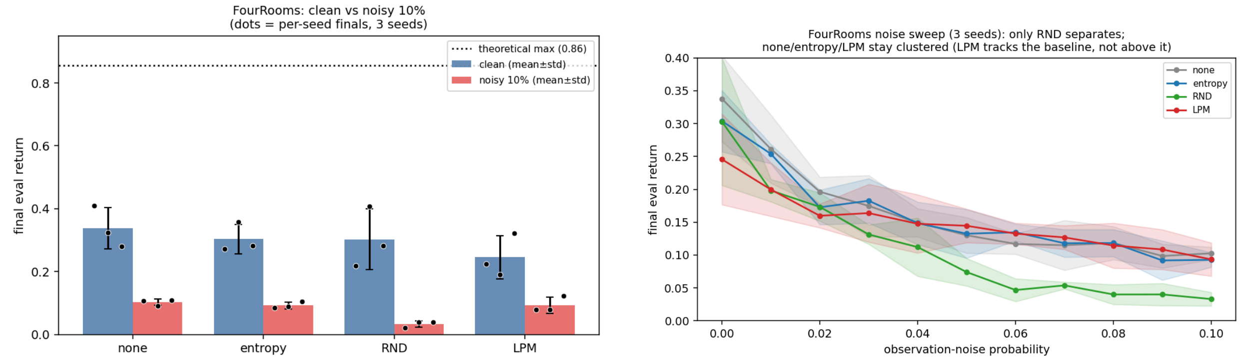
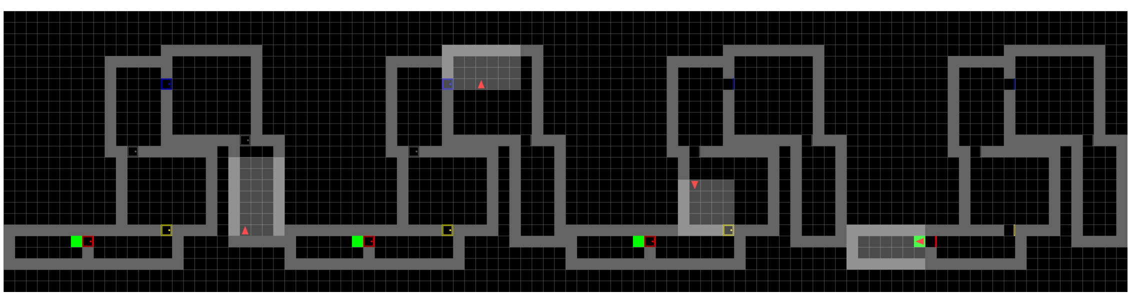
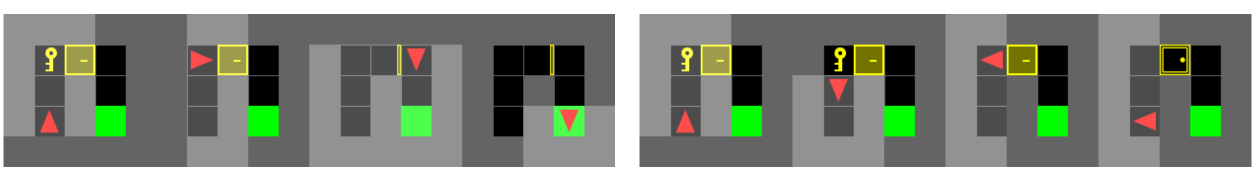


Figure 6: **FourRooms noise robustness** (both panels recomputed on FourRooms; the DoorKey-5x5 numbers are in Table 4). The dotted line in the left panel is the *theoretical-max* eval return for FourRooms (0.86, optimal play; §B) — note how far below it even the clean runs sit. **Left:** clean vs. 10% noise per method (bars = mean±std; dots = per-seed finals, 3 seeds) — every method loses ground, and NONE/ENTROPY/LPM land together ($\approx$ 0.09–0.10) while RND falls lowest (0.03). **Right:** the full sweep `noise_prob` 0–0.1 in steps of 0.01 (bands ±std): **only RND separates** — it degrades fastest and pulls below the pack from $\approx$ 0.04 on ($0.07 \to 0.03$) — while NONE/ENTROPY/LPM stay clustered and statistically tied. LPM *tracks* the baseline (slightly *below* at low noise, its clean under-exploration deficit); it does not beat it.



Hard exploration — RND threads through the MultiRoom-N6 chain to the goal (reward 0.54).



Noise robustness on DoorKey-5x5 (10% noise): LPM (left) solves (LPM is robust on all 8 noisy seeds); RND (right) wanders and fails on a collapse-mode seed — RND-noisy is highly variable (per-seed 0.03–0.96). Representative single episodes.

Figure 7: Trajectory film-strips (evenly-spaced frames). Top: intrinsic motivation enables hard exploration. Bottom: LPM is noise-robust where RND is not.